

Subashree Venkatesan Sundharesan

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Impact driven Software Engineer with 3+ years of experience building scalable fullstack systems, AI-powered applications, and cloud-native platforms. Proven track record of delivering production grade software across enterprise and research environments.

WORK EXPERIENCE

Software Engineer

January 2026 – Present

GlobalNOC

Bloomington, IN, USA

- Engineered real-time **network security & DDoS detection** platform serving **700+ institutions** using **Kafka, Bytewax, OpenSearch, NetFlow** to analyze streaming traffic & trigger automated multi-channel alerts for **24/7** incident response.
- Built **Python** based backend services for multi-cluster **Elastic Stack** environments (Elasticsearch, Logstash, Kibana), adding health checks, **pytest** based unit testing and sparse feed detection to **reduce false positive alerts by 97%**.
- Built a **production-grade logging framework** with structured JSON logging, centralized schemas, cluster-aware context, configurable log levels and noise suppression to standardize logging across distributed services.
- Led migration from **Elasticsearch to open-source OpenSearch**, eliminating licensing costs and delivering **\$1 million+** savings in annual infrastructure while preserving system scalability and performance.

Software Engineer Intern

May 2025 – December 2025

Simons Foundation

New York, NY, USA

- Designed an HPC benchmarks application with an integrated **AI chatbot (OpenAI, RAG, Pinecone)** centralizing 1,000+ runs & **25,000+ data points** from 30+ CPU & GPU architectures, **reducing manual comparison by 80%**.
- Built Python based **ETL pipelines** to parse **10,000+ raw files** (CSV, YAML, JUBE logs), perform unit conversions, handle missing values, and normalize datasets, feeding into a unified **MongoDB** schema that **cut query time by 67%**.
- Developed & deployed **FastAPI** backend services on **AWS** with **sub-second response times** for complex queries, enabling real-time hardware comparisons, performance scaling, anomaly detection, and drill-down diagnostics.
- Built a responsive **React** frontend with **Material UI, Chart.js, D3.js** to deliver **20+ interactive visualizations**, improving decision making for 25+ researchers. Integrated **IP based access control** to secure sensitive results.

Software Development Engineer

March 2022 – August 2024

Amadeus

Bangalore, KA, India

- Developed reusable **Angular** based UI components for Air, Hotel, and Rail booking flows and built scalable **Spring Boot microservices**, supporting **10,000+ production systems** using **Java, Hibernate, Spring Data JPA, JDBC**.
- Implemented sustainability focused features, such as CO2 emissions for flights & eco-friendly hotel recommendations, for environmental awareness. Optimized **Oracle SQL stored procedures**, resulting in **58% faster response time**.
- Achieved **90%+ test case coverage (JUnit, Mockito, Jasmine, Karma)**, ensuring high code quality and reliability, and deployed applications to the **OpenShift** cloud platform via **Jenkins** for quality testing and seamless production rollouts.
- Led **repository migration from BitBucket to GitHub**, migrating repositories and Git history, configuring access controls, and enhancing DevOps workflows using **Microsoft Azure CI/CD**.
- Strengthened security via **re-encryption**, removal of hardcoded credentials, and **CyberArk** based secret management.
- Automated the developer environment setup process using Python, **reducing onboarding time by 75%** (10 to 2.5 days).
- Mentored 7+ engineers** and interns, providing technical guidance and onboarding support to ensure team success.
- Modernized a 20-year legacy monolithic application** by redesigning the frontend with **Angular SPA + Material UI + NgRx**, replacing XSLT, rebuilding the backend using **Spring Boot + PostgreSQL**, replacing Java Servlets, and containerizing with **Docker** for deployment on **Azure App Service**, improving application performance by **33%**.

PROJECTS

Personal AI Assistant  | *AWS Lambda, FastAPI, OpenAI API, GPT-4.1-mini LLM, Pinecone, React, AWS S3, CloudWatch*
Developed a serverless AI Chatbot using a full RAG pipeline with OpenAI GPT-4.1-mini model for reasoning and text-embedding-3-small for semantic search over 100+ uploaded documents, powered by Pinecone vector database. Integrated AWS S3 for automated document ingestion and deployed the FastAPI backend on AWS Lambda, with responsive React chat interface.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C, PL/SQL, Bash/Shell
Backend: Spring Boot, FastAPI, Node.js, Apache Kafka, Flask, Hibernate, JDBC, JUnit, Jest, Pytest, REST APIs
Database: MySQL, Oracle SQL, MongoDB, PostgreSQL, Elasticsearch, OpenSearch, NoSQL, DynamoDB, SQLite, VectorDB
Frontend: HTML, CSS, React, Angular, Tailwind, NgRx, Redux, Material UI, Bootstrap, Chart.js, D3.js
AI/LLM: OpenAI API, Anthropic API, DeepSeek, Llama 3, Pinecone, RAG, Cursor, Claude Code, Codex, GitHub Copilot
Cloud/DevOps: AWS, EC2, Lambda, S3, IAM, CloudWatch, Azure, Jenkins, OpenShift, Docker, Podman
Developer Tools: Git, GitHub, BitBucket, RabbitMQ, Postman, Jira, Unix, Confluence, Agile Methodologies, SAFe, Scrum

EDUCATION

Master of Science in Computer Science

GPA: 3.9/4

Indiana University - Bloomington, IN, USA